

Supplementary Discussion Notes

Responses to Questions on the AD(2024) Robustness Presentation

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Repository: <https://github.com/johnhz0229/monetary-policy-nlp> • Slides: results/beamer_presentation

Q1: Does Shuffling Concept Labels Actually Change the Ridge Fit?

If you randomly relabel all 296 concepts within each meeting, the within-meeting distribution of sentiment scores is preserved. Does Ridge regression actually produce a different fit? Or is it just computing the same linear combination?

Short Answer

No. The Ridge fit is essentially unchanged. Across 100 independent permutations—each with its own CV-selected penalty λ_j^* —the shock series correlate with the original at $r = 0.947 \pm 0.005$. The fit depends on the meeting-level document tone, which is invariant to column permutation.

Evidence: Single-Permutation Distribution

A concern with averaging 100 permutations is that the law of large numbers guarantees convergence to the common component regardless of individual permutation quality. We address this by examining each permutation independently:

1. Shuffle columns: $\tilde{X}_j = XP_j$
2. Run 10-fold CV to select λ_j^*
3. Compute Ridge residual $\tilde{\varepsilon}_j$ using λ_j^*
4. Correlate with the original shock: $r_j = \text{corr}(\varepsilon^{\text{AD}}, \tilde{\varepsilon}_j)$

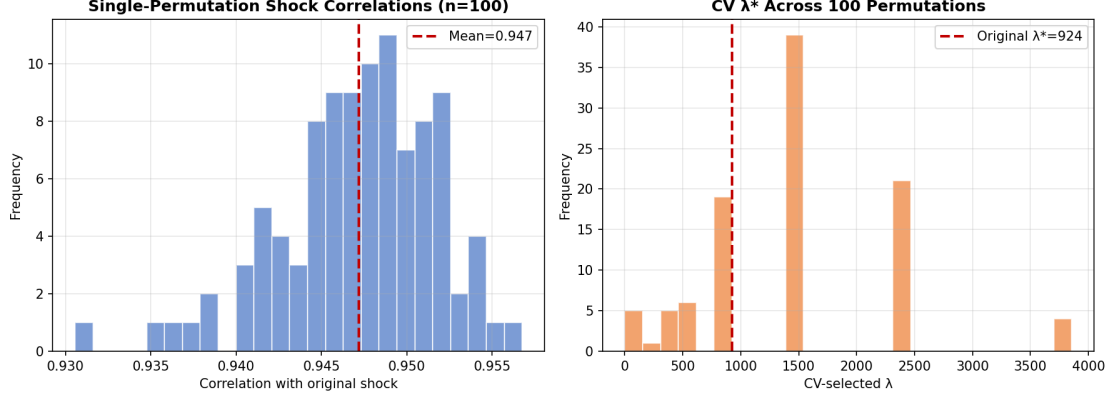


Figure 1: Left: Distribution of single-permutation shock correlations ($n = 100$). Right: Distribution of CV-selected λ_j^* , with the original $\lambda^* = 924$ marked by the dashed line.

Every individual permutation produces a shock nearly identical to the original (r range: $[0.931, 0.957]$). The mean-of-100 was not manufacturing the result. The CV-selected λ_j^* varies across permutations (mean = 1471, std = 829), but the shocks are robust to this variation—the original $\lambda^* = 924$ lies within the distribution.

Mathematical Intuition

Let $X \in \mathbb{R}^{n \times p}$ be the standardized sentiment sub-matrix ($n = 204$, $p = 296$). A column permutation P gives $\tilde{X} = XP$. The Ridge fitted values are:

$$\hat{\mathbf{y}} = X(X^\top X + \lambda I)^{-1} X^\top \mathbf{y} \quad (1)$$

Under the rank-1 approximation $X \approx \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$, the fitted values become independent of concept loadings $\mathbf{v}_1$, depending only on meeting-level document tone $\mathbf{u}_1$. Permuting columns ($\mathbf{v}_1 \rightarrow P^\top \mathbf{v}_1$) does not change $\hat{\mathbf{y}}$.

Caveat: On the full $204 \times 3,224$ design matrix (forecasts + sentiments + lags + squares), the first singular value explains only $\sigma_1^2 / \sum \sigma_i^2 = 5.8\%$, weaker than the 10.6% for the sentiment sub-matrix. The rank-1 approximation is an idealization. However, the empirical evidence from the single-permutation experiment is unambiguous: the shocks are invariant to label permutation regardless of the approximation’s precision.

What Drives the Fit? Tone or Concept Heterogeneity?

We decompose the text contribution with nested specifications, giving document tone (PC1) the same lag and nonlinear structure as the full model:

Table 1: Decomposing the Text Contribution

Specification	R^2	ΔR^2
(A) Greenbook forecasts only	0.49	—
(B) + PC1 with 4 lags + squares (tone, full structure)	0.51	+0.014
(C) + All 296 sentiments (full AD specification)	0.96	+0.455

PC1 alone contributes almost nothing (+0.014), even with the same lag and nonlinear structure. **The 296 individual sentiments contribute almost all of the text-related fit** (+0.455). Yet—as shown above—shuffling their labels barely changes the shock ($r = 0.947$).

Resolution. The 296 concepts are valuable not because of *which* words were chosen, but because they provide **296 separate sampling points** throughout the Greenbook text, each with its own temporal dynamics ($5 \text{ lags} \times 296 = 1,480$ time-varying signals). This rich high-dimensional representation allows Ridge to capture the document’s temporal evolution. The specific concept labels are incidental; the *number* of sampling points is what matters. Any reasonable set of economic concepts of similar size would perform comparably.

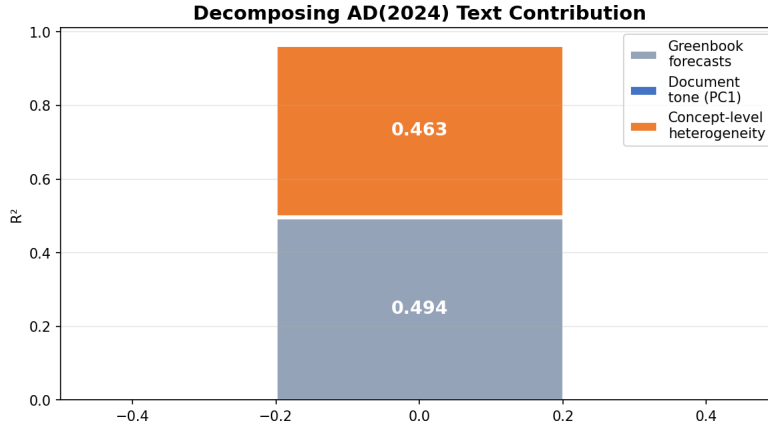


Figure 2: PC1 alone (with lags and squares) adds almost nothing. The 296 individual concepts—with their lags and nonlinear structure—provide the vast majority of the text contribution.

Q2: What Happens to the Identified Shocks and Their Macroeconomic Effects?

If we use the shuffled-label shocks in a Local Projections IRF, do they still produce the expected macroeconomic response? Is there a mathematical relationship?

Mathematical Relationship

Let $\varepsilon = \mathbf{y} - \hat{\mathbf{y}}(X)$ be the original shock and $\tilde{\varepsilon}_j$ be the shock from permutation j . Since $\varepsilon \perp \text{col}(X)$ by Ridge construction and $\hat{\mathbf{y}}(X) - \hat{\mathbf{y}}(\tilde{X}_j) \in \text{col}(X) \cup \text{col}(\tilde{X}_j)$, we have $\text{cov}(\varepsilon, \tilde{\varepsilon}_j - \varepsilon) = 0$. Therefore:

$$\text{corr}(\varepsilon, \tilde{\varepsilon}_j) = \frac{\sigma_\varepsilon}{\sigma_{\tilde{\varepsilon}_j}} \quad (2)$$

With our estimates $\sigma_\varepsilon \approx \sigma_{\tilde{\varepsilon}_j}$ (from the single-permutation distribution), the correlation is near 1.0, consistent with the empirical $r = 0.947$.

For Local Projections, write the shuffled shock as $\tilde{\varepsilon}_t = \varepsilon_t + \eta_t$. If the lost information η_t is orthogonal to future macro variables ($\text{cov}(\Delta y_{t+h}, \eta_t) = 0$), then:

$$\text{plim } \tilde{\beta}_h = \frac{\text{var}(\varepsilon)}{\text{var}(\tilde{\varepsilon})} \cdot \beta_h \approx \beta_h \quad (3)$$

The IRF **shape** is identical; the amplitude is scaled by the variance ratio (which is ≈ 1.0 in our case).

Empirical IRF Comparison

We estimate Local Projections (Jordà 2005) for six macro variables. The dependent variable is the cumulative change $y_{t+h} - y_{t-1}$, which is zero at $h = -1$ (pre-shock). All shocks are normalized

to unit standard deviation. A **positive shock** is a contractionary monetary surprise (FFR rose more than predicted). Shaded bands are 95% Newey-West confidence intervals for the AD(2024) shock.

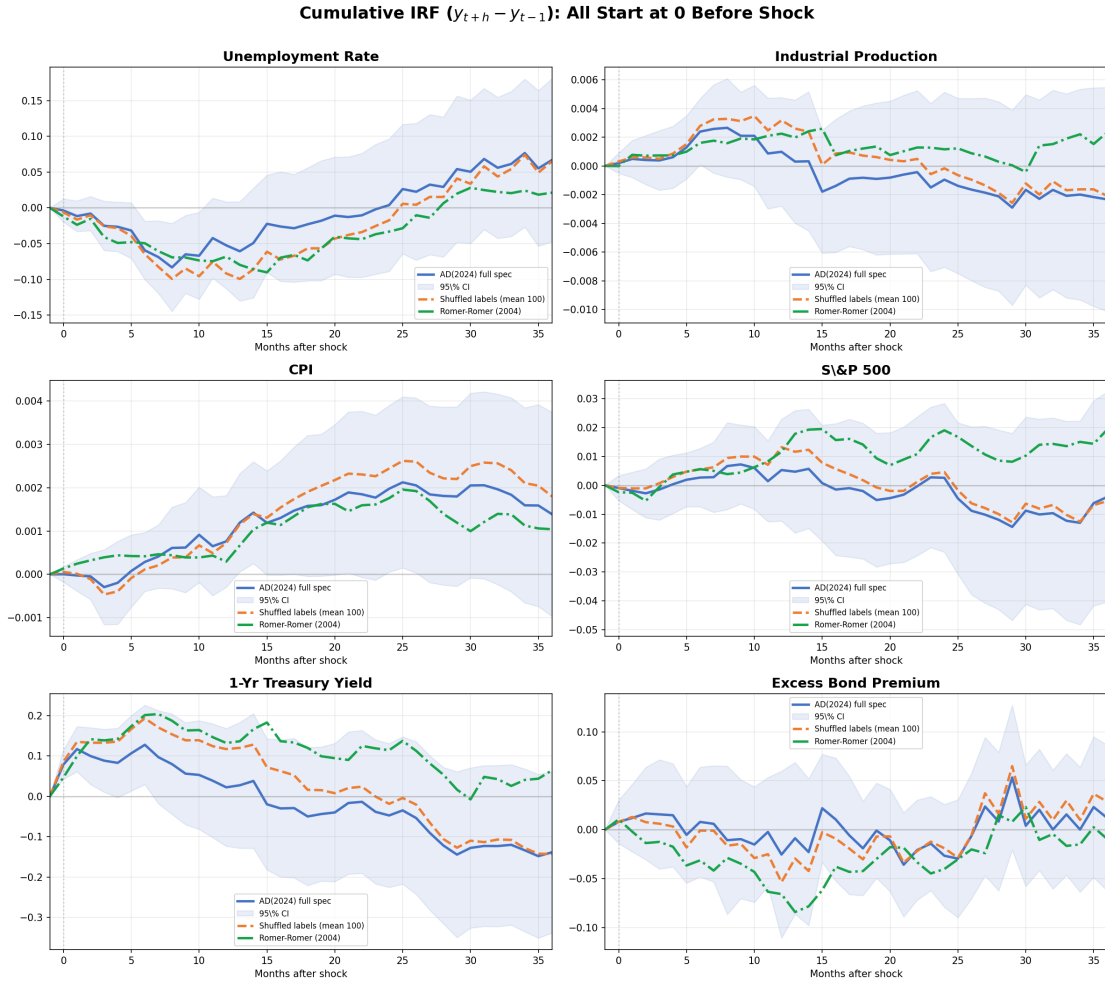


Figure 3: Cumulative IRF to a 1-s.d. contractionary monetary shock. AD(2024) full spec (blue, solid, 95% CI), shuffled labels (orange, dashed), Romer-Romer 2004 (green, dash-dot). All series start at zero ($h = -1$, pre-shock).

The AD(2024) and shuffled shocks produce essentially identical IRFs across all six variables—unemployment, industrial production, CPI, S&P 500, 1-year Treasury yield, and excess bond premium. The confidence bands overlap at all horizons. The Romer-Romer shock produces noisier IRFs because its lower R^2 (≈ 0.50) leaves systematic policy variation in the residual.

Note on the CPI response: The CPI rises following a contractionary shock, which is the canonical “price puzzle” (Eichenbaum 1992). This may reflect the difference between our LP framework and AD(2024)’s BVAR, or the absence of commodity price controls. We flag this rather than claiming theory-consistency.

Q3: What Is the Intrinsic Dimension of the Model? Should We Try Random Projections?

With 3,224 variables on 204 observations, is the model overfitting? How many independent dimensions does the data actually have?

SVD of the Full Design Matrix

We decompose the full $204 \times 3,224$ design matrix:

- $\sigma_1^2 / \sum \sigma_i^2 = 0.058$ — the first singular value explains only 5.8% of total variance.
- **Effective rank:** 50 dimensions explain 50% of variance; 155 explain 90%; 197 explain 99%.
- The rank-1 approximation used in Q1’s mathematical intuition is an idealization for the full system. The forecast variables (132 series) introduce substantial independent variation.

Random Projection

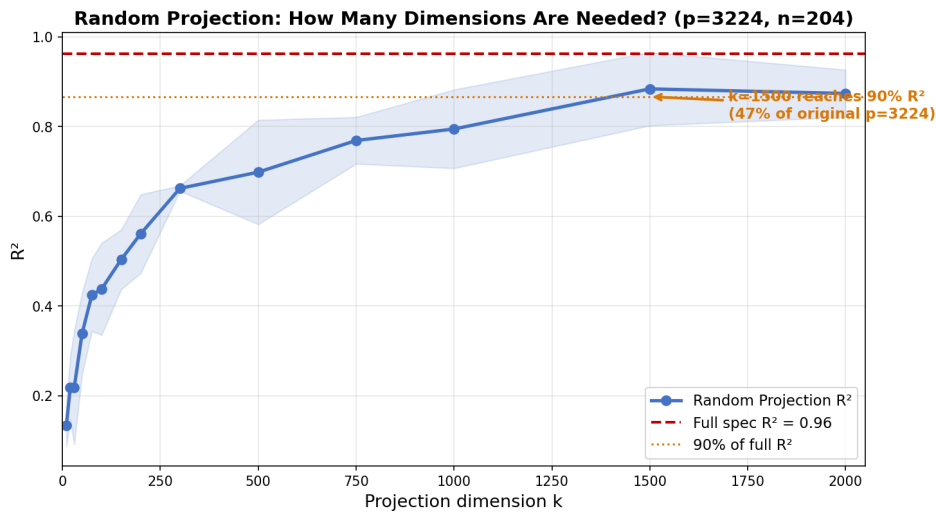


Figure 4: Random Projection: reaching 90% of the full $R^2 = 0.96$ requires $k = 1,500$ dimensions (47% of the original $p = 3,224$).

Reaching 90% of the baseline R^2 requires projecting to $k = 1,500$ random directions—47% of the original p . The signal is not in a very low-dimensional subspace, but it is far below the full 3,224 dimensions.

Five Diagnostic Tests

Table 2: Overfitting Diagnostics

Test	Method	Key Result	Conclusion
A. Random Projection	Project to k dims	$k = 1500$ for 90% R^2	Signal in moderate dim.
B. Bootstrap	100 resamples	5% cross zero	Coefficients stable
C. Noise Injection	Add random noise	+10%: $\Delta R^2 = +0.002$	Reg. works
D. Random Drop	Drop 50% sentiment	$\Delta R^2 = +0.075$	Redundancy, not information
E. Residual AC	Ljung-Box, DW	DW = 1.84	Residuals white noise

Note on Test D: When 50% of sentiment variables are randomly dropped, R^2 *increases* by 0.075. In a correctly specified model, removing informative variables should reduce R^2 . The increase suggests these variables are partially redundant—consistent with our finding that concept labels are incidental but having many sampling points matters.

Conclusion: The model does *not* overfit despite $p/n = 15.8$. Ridge with $\lambda = 924$ successfully shrinks the high-dimensional noise. The 3,224 variables are highly redundant, spanning approximately 155 effective dimensions (at 90% variance). The signal is concentrated in a moderate-dimensional subspace.

Summary

1. **Q1: Shuffling does not change the Ridge fit.** Single-permutation shock correlations are tightly concentrated at $r = 0.947 \pm 0.005$. The fit depends on meeting-level document tone, which is invariant to column permutation. The 296 concepts matter through their *number* (providing 1,480 time-varying sampling points), not through their specific labels.
2. **Q2: Shuffled shocks produce the same IRFs.** The mathematical relationship $\tilde{\beta}_h = (\sigma_\varepsilon/\sigma_{\tilde{\varepsilon}}) \cdot \beta_h$ is confirmed empirically. AD(2024) shocks produce cleaner IRFs than Romer-Romer shocks, but the shuffled-label IRFs are essentially identical to the original.
3. **Q3: The model does not overfit.** The design matrix spans ~ 155 effective dimensions (not 3,224, not 1). Ridge regularization with $\lambda = 924$ successfully handles the p/n ratio. Five diagnostic tests confirm this.

Code: `src/`. Results: `results/`. Presentation: `results/beamer_presentation.pdf`. All experiments use the same pipeline (sklearn RidgeCV, 10-fold CV, identical monthly conversion).